Rishi Raj, PhD

Scientific Computing | High-Performance C++/Python | HPC | Theoretical Physics

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Theoretical physics PhD (Sorbonne University; three peer-reviewed papers) building C++ detector-data compression for CMS at CERN. Combines quantum theory with numerical algorithms and fast C++/Python; trained in FPGA, GPU and OpenMP programming. Seeking engineering roles in scientific software, HPC and quantum computing.

RESEARCH EXPERIENCE

CERN | Research Placement Oct 2026 – present

CMS Next Generation Triggers, Task 3.3 | *Phase-2 RAW data compression*

- Implemented a lossless C++ format for simulated Inner Tracker data: 45% smaller than the original pixel-data representation with the same codec (LZMA-4); exact recovery verified on 400 events.
- Built reproducible codec benchmarks with held-out evaluation; a trained OpenZL compressor reduced simulated Phase-2 detector-data size by 13% versus ROOT ZSTD-4 at matched compression throughput.
- Evaluated compression/reconstruction trade-offs on 2,000 simulated events; preparing a technical note.

Sorbonne University, LPTHE | Doctoral Researcher Oct 2023 – Jun 2026

- Researched BPS black holes and Donaldson-Thomas invariants in string theory, using supersymmetric quantum mechanics, wall-crossing and modularity to study black-hole microstate counting.
- Built symbolic and numerical algorithms for BPS spectra and scattering diagrams in Mathematica; validated reproducible parameter scans against independent formulations.
- Co-authored two papers (JHEP; Annales Henri Poincaré); presented at conferences and seminars.

SELECTED PROJECTS

Holographic Horizons | Numerical Relativity and Scientific ML 2023

Learning Holographic Horizons | *Physical Review D, 2025*

- Solved Einstein equations for non-equilibrium AdS_4 systems with pseudospectral methods; predicted event and apparent horizon areas from 10 engineered time-series features (held-out data, five-fold cross-validation).

High-Performance C++ Simulation | Master's Thesis 2022 – 2023

- Simulated classical bosonic BFSS dynamics (M-theory) in optimized, multithreaded C++17/Boost, approximately 100-200 times faster than a basic Mathematica implementation.
- Studied long-time statistics and thermalization in week-long PBS campaigns on a 64+ core cluster; verified Gauss-law constraints and conservation of energy and angular momentum.

R-Distro | Reproducible ARM64 Linux Build (github.com/rshrj/r-distro) 2026

- Rebuilt Debian's ARM64 base from source at a pinned snapshot as signed, immutable generations; quantified the self-hosting boundary.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, pandas), C++17/20, SQL, Bash, Git, Linux; Mathematica.
Parallel and hardware (coursework): OpenMP, CUDA; VHDL, Vivado/Vitis HLS, Zynq FPGA SoCs.
Modelling and ML: Time series, statistics, cross-validation; PyTorch, scikit-learn, XGBoost.
Computing: ODE/PDE methods, linear algebra, optimization; HPC/PBS, CMake, Docker, pytest.
Languages: English (fluent), French (A1), German (A1).

EDUCATION AND TRAINING

PhD in Theoretical Physics | Sorbonne University 2023 – 2026

Thesis: BPS Black Holes and Donaldson-Thomas Invariants. Defended 10 June 2026.

BS (Hons.) and MS in Theoretical Physics | IIT Madras 2018 – 2023

Minor in Mathematics. MS GPA: 9.59/10; overall GPA: 9.18/10.

CERN STEAM Academy | Scientific Computing and ML Jun – Aug 2026

Grade: 5.5/6. Modern C++, FPGA (VHDL, HLS, SoCs), GPU (CUDA), OpenMP, HPC interconnects, applied ML.

SELECTED AWARDS AND ACTIVITIES

COST STSM grant, University of Geneva (2026); Charpak Lab Fellowship (2023); 63rd and 64th Institute Day Prizes (IIT Madras); KVPY and NIUS Fellowships (2018–2023).

4EU+ PhD School on ML in Physics (2025); CERN ROOT Fixathon (2024) and Patatrack Hackathon (2026).